

Class 35 Cantor's Diagonalization Proofs

Goals

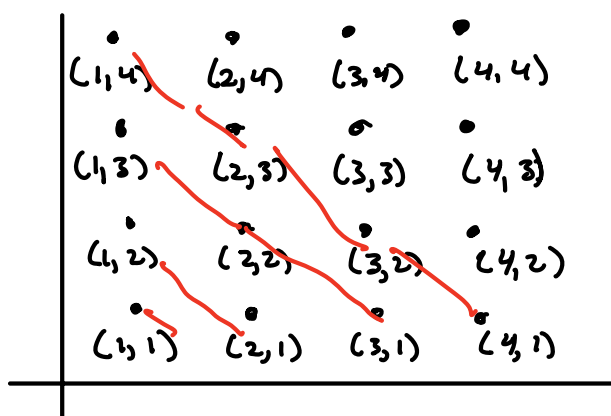
1. Prove the rational numbers are countable.
2. Prove the real numbers are uncountable.
3. Know Cantor's Theorem

Useful Theorems

A countable union of countable sets is countable.

Any subset of a countable set is countable.

Any set that contains an uncountable set is uncountable.



Define a bijection
between the elements
in $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$
by creating a list
of all the points

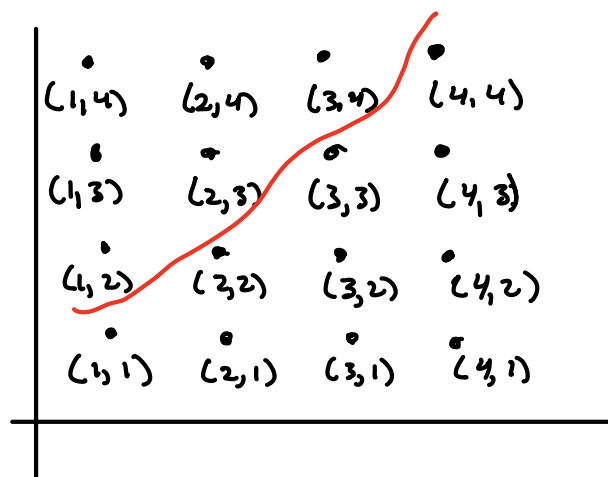
1	2	3	4	5	6	7	8	9	10
(1,1)	(1,2)	(2,1)	(1,3)	(2,2)	(3,1)	(1,4)	(2,3)	(3,2)	(4,1)

This is Cantor's 1st Diagonal method. It
is the key to proving $\mathbb{Q}$ is a countable set.

Let Q_0 be the set of rational numbers in the interval $[0,1)$.

Let Q_1 be rationals in $[1,2)$ and so forth for Q_2 , Q_3 ,

Create a list of elements in $\mathbb{Q}_0$ to prove that the set $\mathbb{Q}_0$ of rational numbers in $[0,1)$ is countable.



1	2	3	4	5	6	7	8	9	10	...
$\frac{1}{2}$	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	...

There is a bijection between $\mathbb{N}$ and the set

$$S = \left\{ \frac{a}{b} : 0 < a < b \text{ with } a, b \in \mathbb{N} \right\}$$

so S is countable.

$\mathbb{Q}_0$ is a subset of S , so $\mathbb{Q}_0$ is countable.

There is a bijection between $\mathbb{Q}_0$ and $\mathbb{Q}_i$ defined by $x \rightarrow i+x$ for each $i \in \mathbb{Z}$

so each $\mathbb{Q}_i$ is countable.

$\mathbb{Q} = \bigcup_{i \in \mathbb{Z}} \mathbb{Q}_i$ is the union of a countable number of countable sets, so $\mathbb{Q}$ is countable.

Cantor's second diagonal method is the key to proving the set of real numbers is uncountable.

We will prove that the open interval $(0,1)$ is uncountable. This forces us to conclude the entire set of real numbers is uncountable since it contains an uncountable subset.

Assume for contradiction that $(0, 1)$ is a countable set.

This means we could construct a list of all the real numbers in $(0,1)$.

$$a_1 = 0.a_{11} a_{12} a_{13} a_{14} \dots$$

$$a_2 = 0.a_{21} a_{22} a_{23} a_{24} \dots$$

$$a_3 = 0.a_{31} a_{32} a_{33} a_{34} \dots$$

$$a_4 = 0.a_{41} a_{42} a_{43} a_{44} \dots$$

$\vdots$

Cantor argued that this cannot be a complete list by constructing a real number in $[0,1)$ that is different from every a_i in the list.

$$b = 0.b_1 b_2 b_3 b_4 \dots \quad \text{where } b_i = 1 \text{ if } a_{ii} = 0$$

$$\text{and } b_i = 2 \text{ if } a_{ii} \neq 0$$

So b cannot be any a_i in the list since b and a_i are different in the i^{th} decimal place.

Cantor's Theorem: If A is any set then, the cardinality of A is strictly less than the cardinality of the power set of A .

Proof:

Define a function f with domain A and codomain $P(A)$ as follows.

For any $x \in A$, $f(x) = \{x\}$ in $P(A)$

This function is one-to-one since

If $f(x) = f(y)$ we have $\{x\} = \{y\}$ so $x = y$.

Therefore $|A| \leq |P(A)|$.

Now prove there is no bijection between A and $P(A)$ by contradiction.

Assume for contradiction there is a bijection g , with domain A mapping onto all of $P(A)$.

Cantor constructed a set D in $P(A)$ that is not in the range of g , contradicting g mapping onto $P(A)$.

**** Define $D = \{x \in A : x \notin g(x)\}$.**

Now $D \subseteq A$ so $D \in P(A)$, but there is no $d \in A$ such that $g(d) = D$.

There are only 2 possibilities for d .

Either ① $d \in D$ or ② $d \notin D$. Both possibilities lead to trouble.

① If $d \in D$, then $d \notin g(d)$ by defn of D
But $g(d) = D$ so $d \in D$ and $d \notin D$ (X)

② If $d \notin D$, then $d \in g(d) = D$ by defn of D .
Again $d \notin D$ and $d \in D$ (X)

This means there is no element in A that g will map to the set D .

So g is not onto all of $P(A)$. So g cannot be a bijection from A to $P(A)$.

Therefore $|A| \neq |P(A)|$ meaning $|A|$ is strictly less than $|P(A)|$.

Example Let $A = \mathbb{N}$. Suppose $g: \mathbb{N} \rightarrow P(\mathbb{N})$

starts like

$$\begin{aligned} g(1) &= \\ g(2) &= \\ g(3) &= \\ g(4) &= \\ g(5) &= \\ g(6) &= \end{aligned}$$

Construct $D = \{$

Is x in $g(x)$? If yes, then $x \notin D$.

If no, then $x \in D$